

Lagrangian Lower Preservation for Laminar Cut-Constrained Spanning Trees

Corrected Reduction and a Dual-Aligned Partial Result

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Abstract

We revisit the open question, highlighted by Nägele and Zenklusen, of whether laminar cut-constrained minimum spanning trees admit a polynomial-time constant multiplicative violation while preserving the exact optimum cost. The conclusion of this note is deliberately modest: we do *not* obtain a solution of the open problem. We correct a proposed Lagrangian argument by replacing an unjustified equivalence with a one-way certificate theorem. We then prove a new partial result: if an optimal primal-dual LP pair is aligned inside the matroid of Lagrangian-minimum spanning trees, then a polynomial-time algorithm returns a tree of cost at most the exact constrained optimum and with multiplicative violation at most 3. This includes, as a special case, the previously identified aligned-LP situation. The proof combines LP duality, the matroid of minimum-weight bases, and the aligned laminar-basis theorem of Klein and Olver.

Status. The results below are partial progress, not a resolution of the general open problem. The original draft's strongest claim, that the problem is equivalent to a penalty-balanced rounding lemma, is not justified. The rigorous statement is one-way: such a rounding theorem would suffice. The unconditional theorem proved here applies only under an additional dual-alignment hypothesis.

1 Problem statement

Let $G = (V, E)$ be a connected undirected graph, let $c \in \mathbb{R}_{\geq 0}^E$, and let $\mathcal{L} \subseteq 2^V \setminus \{\emptyset, V\}$ be laminar. For a nonempty proper set $S \subset V$, write

$$\delta(S) = \{uv \in E : |\{u, v\} \cap S| = 1\}.$$

For each $S \in \mathcal{L}$ an integer upper bound $b_S \in \mathbb{Z}_{\geq 0}$ is given. The exact constrained optimum is

$$\text{OPT} = \min\{c(T) : T \text{ is a spanning tree of } G, |T \cap \delta(S)| \leq b_S \text{ for all } S \in \mathcal{L}\}.$$

The open question asks whether there are a universal constant β and a polynomial-time algorithm which, whenever the exact problem is feasible, returns a spanning tree T satisfying

$$c(T) \leq \text{OPT}, \quad |T \cap \delta(S)| \leq \beta b_S \quad \forall S \in \mathcal{L}.$$

The exact-cost requirement is the difficult part. Known general-laminar thin-tree algorithms give a constant multiplicative violation with a constant cost factor, while iterative-relaxation algorithms give exact cost with additive rather than multiplicative violation.

Let M_0 denote the graphic matroid of G , and let $P_{\text{ST}}(G) = P(M_0)$ be its base polytope. Define $A \in \{0, 1\}^{\mathcal{L} \times E}$ by

$$A_{S,e} = \mathbf{1}[e \in \delta(S)].$$

The natural relaxation is

$$\nu = \min\{c^\top x : x \in P_{\text{ST}}(G), Ax \leq b\}. \quad (1)$$

Every exact feasible tree is feasible for this relaxation, so $\nu \leq \text{OPT}$.

2 The corrected Lagrangian certificate

The following statement is the part of the original Lagrangian idea that is correct and useful. It is a sufficient certificate, not an equivalence.

Lemma 2.1 (Primal-dual Lagrangian certificate). *Let x^* be an optimal solution of (1). There is a vector $y \in \mathbb{R}_{\geq 0}^{\mathcal{L}}$ such that, for*

$$d = c + A^\top y,$$

x^ minimizes $d^\top z$ over $P_{\text{ST}}(G)$, and $y_S((Ax^*)_S - b_S) = 0$ for every $S \in \mathcal{L}$. Consequently, if a spanning tree T satisfies*

$$d(T) = d^\top x^* \quad (2)$$

and

$$y^\top A\chi_T \geq y^\top Ax^*, \quad (3)$$

then $c(T) \leq c^\top x^$.*

Proof. The first assertion is the KKT optimality condition for minimizing $c^\top x$ over $P_{\text{ST}}(G) \cap \{x : Ax \leq b\}$. Equivalently, after dualizing the side constraints $Ax \leq b$, there is $y \geq 0$ such that x^* minimizes $(c + A^\top y)^\top z$ over the base polytope, with complementary slackness on the side constraints.

For the final implication, put $w = A^\top y$. Then $d = c + w$. If (2) holds, then

$$c(T) + w(T) = d(T) = d^\top x^* = c^\top x^* + w^\top x^*.$$

Since $w(T) = y^\top A\chi_T$ and $w^\top x^* = y^\top Ax^*$, condition (3) gives

$$c(T) = c^\top x^* + y^\top Ax^* - y^\top A\chi_T \leq c^\top x^*.$$

□

Theorem 2.2 (One-way reduction). *Suppose there is a polynomial-time procedure which, for every optimal primal-dual pair (x^*, y) as in Lemma 2.1, returns a spanning tree T satisfying*

$$d(T) = d^\top x^*, \quad y^\top A\chi_T \geq y^\top Ax^*, \quad |T \cap \delta(S)| \leq \beta b_S \quad \forall S \in \mathcal{L}.$$

Then the open problem has a polynomial-time β -multiplicative exact-cost algorithm.

Proof. The cut guarantees are exactly the desired multiplicative guarantees. By Lemma 2.1, $c(T) \leq c^\top x^*$. Since x^* is feasible for the LP relaxation, $c^\top x^* = \nu \leq \text{OPT}$. □

Remark 2.3 (What was wrong with the stronger equivalence claim). A tree satisfying the original open problem need not lie in the minimal face of $P_{\text{ST}}(G)$ containing a chosen LP optimum x^* , and it need not satisfy the weighted lower inequality (3) for a chosen dual multiplier y . These requirements are therefore additional sufficient conditions, not a reformulation of the original problem. The algebra above explains why they are natural: they are exactly what makes the Lagrangian identity certify cost factor one.

3 The optimal-face matroid

We next isolate a condition under which the missing weighted lower inequality can be forced by an existing aligned laminar-rounding theorem.

Let M be a matroid on ground set E with base family $\mathcal{B}(M)$ and base polytope $P(M)$. For a weight vector $d \in \mathbb{R}^E$, define

$$\mathcal{B}_d(M) = \arg \min \{d(B) : B \in \mathcal{B}(M)\}.$$

Lemma 3.1 (Minimum-weight bases form a matroid). *The family $\mathcal{B}_d(M)$ is the family of bases of a matroid, denoted M_d . Moreover,*

$$P(M_d) = \{x \in P(M) : d^\top x = \min_{z \in P(M)} d^\top z\}.$$

If M is graphic and d is given explicitly, then M_d has a polynomial-time rank oracle.

Proof. Let $\lambda_1 < \dots < \lambda_k$ be the distinct values of d , and let $F_i = \{e \in E : d_e \leq \lambda_i\}$, with $F_0 = \emptyset$. The greedy algorithm for matroids implies that a base B has minimum d -weight if and only if

$$|B \cap F_i| = r_M(F_i) \quad \text{for every } i = 1, \dots, k.$$

Equivalently, the elements chosen from each layer $F_i \setminus F_{i-1}$ form a base of the minor $(M|F_i)/F_{i-1}$. Hence the minimum-weight bases are exactly the bases of the direct sum

$$M_d = \bigoplus_{i=1}^k (M|F_i)/F_{i-1}.$$

This proves the matroid claim. The equality of base polytopes follows because the face of a matroid base polytope exposed by d is the convex hull of the d -minimum bases. For a graphic matroid, ranks of restrictions and contractions are computable by graph connectivity, and the displayed direct-sum representation gives a polynomial-time rank oracle. $\square$

For $S \subseteq V$, write

$$E_{\text{in}}(S) = \{uv \in E : u, v \in S\}$$

for the internal edge set induced by S .

Definition 3.2 (Point alignment in a matroid). Let M be a matroid on E and let $x \in P(M)$. We say that x is $\mathcal{L}$ -aligned in M if

$$x(E_{\text{in}}(S)) = r_M(E_{\text{in}}(S)) \quad \forall S \in \mathcal{L}.$$

We say that M is $\mathcal{L}$ -aligned if every point of $P(M)$ is $\mathcal{L}$ -aligned, equivalently if every basis B of M satisfies $|B \cap E_{\text{in}}(S)| = r_M(E_{\text{in}}(S))$ for every $S \in \mathcal{L}$.

We will use the following theorem of Klein and Olver.

Theorem 3.3 (Klein–Olver aligned laminar-basis theorem). *Let $G = (V, E)$ be a graph, let M be a matroid on E that is $\mathcal{L}$ -aligned, and let $b \in \mathbb{Z}_{\geq 0}^E$. Let $x^0 \in P(M)$ satisfy $x^0(\delta(S)) \leq b_S$ for every $S \in \mathcal{L}$. Then for every nonnegative cost vector $a \in \mathbb{R}_{\geq 0}^E$ one can find in polynomial time a basis B of M such that*

$$a(B) \leq a^\top x^0, \quad |B \cap \delta(S)| \leq 2b_S + 1 \quad \forall S \in \mathcal{L}.$$

Remark 3.4. The theorem is stated here in the form used below. Klein and Olver’s Theorem 4.6 solves the aligned-matroid relaxation with cost a and returns a basis whose cost is at most the LP optimum; since x^0 is feasible, that cost is at most $a^\top x^0$. Their theorem also gives the displayed $2b_S + 1$ cut bounds.

We also use the standard refinement observation from the same work.

Lemma 3.5 (Refining an aligned point). *If $x \in P(M)$ is $\mathcal{L}$ -aligned in M , then by repeatedly refining M with respect to the sets $E_{\text{in}}(S)$, $S \in \mathcal{L}$, one obtains a matroid M' such that*

$$x \in P(M'), \quad P(M') \subseteq P(M), \quad M' \text{ is } \mathcal{L}\text{-aligned}.$$

If M has a polynomial-time rank oracle, so does M' .

Proof. Refinement with respect to a set R means replacing M by the direct sum of $M|_R$ and M/R . If $x(R) = r_M(R)$, then x remains in the base polytope after this refinement. Because the sets $E_{\text{in}}(S)$ form a laminar family of edge sets when $\mathcal{L}$ is laminar, refining successively over all $S \in \mathcal{L}$ makes each such set tight for every basis of the resulting matroid. This is exactly $\mathcal{L}$ -alignment. Rank oracles are preserved by restriction, contraction, and direct sum. $\square$

4 A dual-aligned exact-cost partial result

We now prove the main partial theorem. It is the principal repair of the original draft: the missing lower-preservation inequality is obtained by running the aligned laminar-basis theorem with a carefully chosen auxiliary cost.

Definition 4.1 (Dual-aligned optimal face). Let (x^*, y) be an optimal primal-dual pair for (1), and put $d = c + A^\top y$. Let M_d be the matroid whose bases are the d -minimum spanning trees of G . We say that (x^*, y) is *dual-aligned* if x^* is $\mathcal{L}$ -aligned in M_d , i.e.

$$x^*(E_{\text{in}}(S)) = r_{M_d}(E_{\text{in}}(S)) \quad \forall S \in \mathcal{L}.$$

Theorem 4.2 (Exact cost and 3-violation under dual alignment). *Assume the exact laminar-constrained instance is feasible. Suppose we are given an optimal primal-dual pair (x^*, y) for (1) that is dual-aligned in the sense of Definition 4.1. Then one can find in polynomial time a spanning tree T such that*

$$c(T) \leq \text{OPT}, \quad |T \cap \delta(S)| \leq 3b_S \quad \forall S \in \mathcal{L}.$$

In fact, the stronger additive-multiplicative bound $|T \cap \delta(S)| \leq 2b_S + 1$ holds for every $S \in \mathcal{L}$.

Proof. Let $d = c + A^\top y$ and let M_d be the matroid of d -minimum spanning trees. By Lemma 2.1, $x^* \in P(M_d)$. By the dual-alignment assumption and Lemma 3.5, refine M_d along the sets $E_{\text{in}}(S)$ to obtain an $\mathcal{L}$ -aligned matroid M' such that

$$x^* \in P(M') \subseteq P(M_d).$$

Thus every basis of M' is a d -minimum spanning tree of G .

Put

$$w = A^\top y, \quad \Gamma = \max_{e \in E} w_e, \quad q_e = \Gamma - w_e \quad (e \in E).$$

Then $q \geq 0$. Apply Theorem 3.3 to the aligned matroid M' , the laminar bounds b , the feasible point x^* , and the auxiliary cost q . We obtain a basis T of M' such that

$$q(T) \leq q^\top x^*, \quad |T \cap \delta(S)| \leq 2b_S + 1 \quad \forall S \in \mathcal{L}.$$

Since T is a basis of $M' \subseteq M_d$, it is a d -minimum spanning tree, so

$$d(T) = d^\top x^*.$$

All bases of M' have cardinality $|V| - 1$, and $x^*(E) = |V| - 1$. Hence

$$q(T) = \Gamma(|V| - 1) - w(T), \quad q^\top x^* = \Gamma(|V| - 1) - w^\top x^*.$$

The inequality $q(T) \leq q^\top x^*$ is therefore equivalent to

$$w(T) \geq w^\top x^*.$$

Since $w = A^\top y$, this is exactly

$$y^\top A \chi_T \geq y^\top A x^*.$$

Lemma 2.1 now gives

$$c(T) \leq c^\top x^* \leq \text{OPT}.$$

Finally, because the exact instance is feasible, every nonempty proper cut $\delta(S)$ with $S \in \mathcal{L}$ is crossed at least once by any feasible spanning tree. Therefore $b_S \geq 1$ for all $S \in \mathcal{L}$, and

$$2b_S + 1 \leq 3b_S.$$

This proves the claimed multiplicative guarantee. $\square$

Remark 4.3 (Checkability). For a fixed optimal primal-dual pair (x^*, y) , the hypothesis of Theorem 4.2 is polynomial-time checkable: construct the minimum-base matroid M_d using Lemma 3.1, compute $r_{M_d}(E_{\text{in}}(S))$ for all $S \in \mathcal{L}$, and compare with $x^*(E_{\text{in}}(S))$. The theorem should therefore be viewed as a certified partial algorithm: when this check succeeds, the conclusion follows. I do not prove that every feasible instance admits such a pair.

5 The aligned-LP case as a corollary

The original draft contained a correct but limited aligned-case observation. Theorem 4.2 recovers it as a corollary, and explains why the Lagrangian lower condition can be enforced in that case.

Corollary 5.1 (Aligned optimum in the original graphic matroid). *Assume the exact instance is feasible. Suppose (1) has an optimal solution x^* satisfying*

$$x^*(E_{\text{in}}(S)) = r_{M_0}(E_{\text{in}}(S)) \quad \forall S \in \mathcal{L},$$

where M_0 is the graphic matroid of G . Then one can find in polynomial time a spanning tree T with

$$c(T) \leq \text{OPT}, \quad |T \cap \delta(S)| \leq 3b_S \quad \forall S \in \mathcal{L}.$$

Proof. Choose dual multipliers y as in Lemma 2.1, put $d = c + A^\top y$, and let M_d be the d -minimum-base matroid. Since x^* minimizes d over $P(M_0)$, we have $x^* \in P(M_d)$. For every $S \in \mathcal{L}$,

$$x^*(E_{\text{in}}(S)) \leq r_{M_d}(E_{\text{in}}(S)) \leq r_{M_0}(E_{\text{in}}(S)).$$

The first inequality is a rank inequality in the base polytope $P(M_d)$; the second follows because every independent set of M_d is independent in M_0 . By the assumed graphic alignment, the left and right sides are equal. Hence x^* is $\mathcal{L}$ -aligned in M_d . Theorem 4.2 applies. $\square$

Remark 5.2 (Algorithmic detection of the corollary’s hypothesis). The existence of a graphic-aligned optimal LP solution can be tested by first computing the optimum value ν of (1), then resolving the LP with the additional equalities $x(E_{\text{in}}(S)) = r_{M_0}(E_{\text{in}}(S))$ for all $S \in \mathcal{L}$ and objective value forced to be ν . If feasible, the resulting point satisfies the corollary’s hypothesis.

6 What remains open

The proof of Theorem 4.2 suggests a precise way to strengthen the partial result. Given an arbitrary optimal primal-dual pair (x^*, y) , it would suffice to find a d -minimum spanning tree T such that

$$|T \cap \delta(S)| \leq O(1)b_S \quad \forall S \in \mathcal{L}, \quad y^\top A\chi_T \geq y^\top Ax^*.$$

Theorem 4.2 proves this when the d -minimum face can be refined into an aligned matroid containing x^* . The general case is exactly where the current proof stops: without alignment, the Klein–Olver aligned-matroid theorem cannot be applied with the auxiliary cost $q = \Gamma \mathbf{1} - A^\top y$.

The following statement is therefore a sufficient target, but it is not known from the arguments here.

Conjecture 6.1 (Sufficient penalty-balanced optimal-face rounding). *There is a universal constant β and a polynomial-time algorithm which, for every optimal primal-dual pair (x^*, y) of (1), with $d = c + A^\top y$, returns a d -minimum spanning tree T satisfying*

$$|T \cap \delta(S)| \leq \beta b_S \quad \forall S \in \mathcal{L}, \quad y^\top A\chi_T \geq y^\top Ax^*.$$

By Theorem 2.2, Conjecture 6.1 would solve the open problem. Notice that this conjecture is weaker and more accurately targeted than the original draft’s formulation: it does not require the tree to lie in the minimal face containing x^* , and it measures upper violation against the actual bounds b_S rather than against every coordinate of Ax^* .

7 Verification summary

The unconditional claims in this note have the following dependency chain.

1. Lemma 2.1 is standard LP duality/KKT for the natural spanning -tree relaxation.
2. Lemma 3.1 is the standard greedy characterization of minimum-weight matroid bases.
3. Lemma 3.5 is the standard matroid refinement step used by Klein and Olver to pass from an aligned point to an aligned matroid.
4. Theorem 4.2 uses Klein–Olver’s aligned laminar-basis theorem once, with the auxiliary non-negative cost $q = \Gamma \mathbf{1} - A^\top y$. The inequality $q(T) \leq q^\top x^*$ is exactly the missing dual-weighted lower inequality.

Thus the note proves a genuine partial result, but not a full solution. The remaining obstacle is to obtain the same weighted lower preservation and constant laminar upper bounds without the dual-alignment hypothesis.

References

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